BLADES PLC CASE STUDIES

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Blades 6. Assessment of Purchasing Power Parity

1. What is the relationship between the exchange rates and relative inflation levels of the two countries? How will this relationship affect Blades' Thai revenue and costs given that the baht is freely floating? What is the net effect of this relationship on Blades?

According to the theory of Purchasing Power Parity (PPP), exchange rates between two countries should adjust to reflect the difference in their inflation rates. In this case, if Thailand experiences a higher inflation rate than the UK, the Thai baht is expected to depreciate against the British pound. For a company like Blades, this would mean that when it converts its Thai baht revenue back into pounds, the amount received would decrease.

If Blades imports components from Thailand and pays for them in baht, a weaker baht would initially make these imports cheaper when converted into pounds. However, higher inflation in Thailand could drive up the prices of those components, potentially canceling out the benefit of the baht's depreciation.

If the baht depreciates due to higher inflation in Thailand, Blades' revenue in pounds might take a hit. However, this could be partially offset by reduced import costs. The net effect will depend on how much the exchange rate changes compared to the inflation gap between the two countries.

2. What are some of the factors that prevent PPP from occurring in the short run? Would you expect PPP to hold better if countries negotiate trade arrangements under which they commit themselves to the purchase or sale of a fixed number of goods over a specified time period? Why or why not?

In short run, transportation costs can prevent PPP from occurring by causing price disparities between conntries and tariffs and trade barries might cause this as well because taxex, import quotas, and other restrictions disrupt price equality. Other factors like market imperfections and exchange rate volatilities could alse lead to the same result

If countries negotiate trade agreements involving fixed numbers of goods over specified periods, PPP may hold better. This is because such agreements can reduce the impact of supply and demand fluctuations, stabilize prices, and provide a predictable exchange rate environment. However, this approach doesn't completely eliminate factors like transportation costs and tariffs.

3. How do you reconcile the high level of interest rates in Thailand with the expected change of the baht-pound exchange rate according to PPP?

High interest rates in Thailand typically attract foreign capital, leading to an appreciation of the baht. However, according to PPP, if Thailand has a higher inflation rate than the UK, the baht should depreciate in the long run. This can be reconciled by considering the relationship between interest rates, inflation, and exchange rates:

Interest Rate Parity (IRP): Suggests that the difference in nominal interest rates between two countries is equal to the expected change in exchange rates.

PPP: Indicates that exchange rates adjust to the inflation differential.

In the short term, high interest rates may attract foreign investment, causing the baht to appreciate. Over time, if Thailand's inflation is consistently higher, the PPP effect would lead to a depreciation of the baht to reflect the relative changes in price levels.

4. Given Blades' future plans in Thailand, should the company be concerned with PPP? Why or why not?

Blades should be concerned with PPP. If Blades plans to continue exporting and importing in Thailand over the long term, changes in exchange rates due to differing inflation levels could significantly impact both its revenue and costs. As PPP predicts that exchange rates will adjust to inflation differences between countries, Blades must be aware of these potential fluctuations.

If inflation in Thailand remains high, the baht is likely to depreciate, reducing the pound-denominated value of Blades' Thai revenue. Conversely, the cost of importing raw materials from Thailand could decrease if the baht weakens. Understanding and monitoring these dynamics is crucial for long-term planning and risk management.

5. PPP may hold better for some countries than for others. Given that the Thai baht has been freely floating for only a short period of time, how do you think Blades can gain insight into whether PPP will hold for Thailand?

To examine whether PPP will hold for Thailand, Blades can analyze the historical data, i.e. the past exchage rate and inflation data to identify whether deviations from PPP have been temporary or persistent. Secondly, a benchmark can be used, for example, to compare with other countries who has the similar economics with freely floating currencies and observe how well PPP holds in those countries. Additionally, Blades can also consult financial experts to obtain insights from local financial analysits and international economists specializing in emerging markets.

Blades 7. Assessment of potential arbitrage opportunities

1. Determine whether the foreign exchange quatations are appropriate. If not, determine the profit you could generate by withdrawing £100 000 from Blades' cheque account and engageing in arbitrage before the rates are adjusted

	Minzu Bank	Sobat Bank
Bid	£0.0149	£0.0152
Ask	£0.0151	£0.0153

Table 1: Spot rate quotations from two banks

Identify the arbitrage opportunity

Minzu Bank is willing to sell Thai baht at £0.0151 (ask price). **Sobat Bank** is willing to buy Thai baht at £0.0152 (bid price).Since the selling price at Minzu Bank (£0.0151) is lower than the buying price at Sobat Bank (£0.0152), there is an opportunity for locational arbitrage.

Buy Thai baht at Minzu Bank's ask price

We start with £100,000. Ask price at Minzu Bank: £0.0151/THB. So the amount of THB we can purchase:

$$\text{THB} = \frac{100,000}{0.0151} \approx 6,622,516.56\ \text{THB}.$$

Sell Thai baht at Sobat Bank's bid price

Bid price at Sobat Bank: £0.0152/THB. Amount we will receive in GBP:

$$\text{GBP} = 6,622,516,56 \times 0.0152 \approx 100,662.25\ \text{GBP}.$$

The profit

$$\text{profit:} = 100,662.25 - 1,000,000 = 662.25\ \text{GBP}.$$

2. Besides the bid and ask quotes for the Thai baht, Minzu Bank has provided the following quotations for the pound and the Japanese yen. Determine whether the cross exchange rate between the Thai baht and Japanese yen is appropriate. If not, determine the profit you could generate for Blades by withdrawing £100 000 from Blades' current account and engaging in triangular arbitrage before the rates are adjusted.

	Quoted Bid Rate	Quoted Ask Rate
Value of a Japanese yen in British pounds	£0.0057	£0.0058
Value of a Thai baht in Japanese yen	¥2.69	¥2.70

Table 2: Quotations for the pound and the japanese yen

Implied Cross-Rate (THB/JPY), initially £100,000

$$\text{JPY} = \frac{100,000}{0.0058} \approx 17,241,379.31\ \text{JPY}$$

JPY to THB

$$\text{THB} = \frac{17,241,379.31}{2.70} \approx 6,385,696.04\ \text{THB}$$

Implied cross-rate: THB/JPY exchange rate

$$\text{Implied rate} = \frac{6,385,696.04}{17,241,379.31} \approx 0.3704$$

The quoted rate is ¥2.70 / THB, which implies the reciprocal should be approximately 0.3704. Hence, the cross exchange rate between the Thai baht and the Japnese yen is appropriate.

3. Determine whether the forward rate is priced appropriately. If not, calculate the potential profit for Blades by engaging in covered interest arbitrage.

Given:

- Spot rate: £0.0151/THB
- 90-day forward rate: £0.015/THB
- UK interest rate: 2% annual (0.5% for 90 days)
- Thailand interest rate: 3.75% annual (0.9375% for 90 days)
- Initial investment: £100,000

Theoretical forward rate using IRP.

$$F = S \times \frac{(1 + r_{\text{domestic}})}{(1 + r_{\text{foreign}})},$$

where:

- $S = 0.0151$,
- $r_{\text{domestic}} = 0.005$ $(UK\ 90 - day\ rate)$,
- $r_{\text{foreign}} = 0.009375$ $(Thailand\ 90 - day\ rate)$.

$$F = 0.0151 \times \frac{1 + 0.005}{1 + 0.009375} \approx 0.015$$

The quoted forward rate (£0.015) matches the calculated forward rate, indicating that it is priced appropriately. Therefore, covered interest arbitrage is not possible. Blades cannot generate a profit from engaging in covered interest arbitrage in this scenario.

4. Why are arbitrage opportunities likely to disappear soon after they have been discovered? How the baht's spot and forward rates would adjust until covered interest arbitrage is no longer possible. What is the resulting equilibrium state called?

When traders identify an arbitrage opportunity, they quickly buy or sell the relevant assets (currencies, securities) to capitalize on the price differences. This action changes the supply and demand dynamics in the market.

If arbitrageurs start buying the Thai baht in the spot market (to profit from the interest rate differential between Thailand and the UK), the increased demand for the baht will cause the spot rate of the baht to appreciate.Simultaneously, arbitrageurs would sell the baht in the forward market (locking in future profits). This increased supply of baht in the forward market puts downward pressure on the forward rate, causing it to depreciate.

The arbitrage process continues until the difference between the spot rate, forward rate, and interest rate differentials reaches equilibrium. This equilibrium state is known as Interest Rate Parity (IRP)